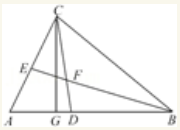
Rollouts Analysis on Process (I)



Problem:
In the diagram, point F is the intersection of the angle bisectors CD and BE of triangle ABC, and CG is perpendicular to AB at point G. If angle ACG is 32 degrees, what is the measure of angle BFC?

Perfect Rollout:

Step 1: Recognize that CD and BE are angle bisectors, so point F is the incenter of triangle ABC.

Step 2: Since CG is perpendicular to AB, triangle CGA is a right triangle.

Step 3: In triangle CGA, using the fact that the sum of angles is 180°, calculate angle CAG: angle CAG = 90° - angle ACG =90° - 32° = 58°.

Step 4: Since CD is the angle bisector of angle ACB, angle ACD = angle BCD = 1/2 angle ACB.

Step 5: Let angle ACB = 2θ. Then angle ACD = angle BCD = θ.

Step 6: Similarly, BE is the angle bisector of angle ABC, so angle ABE = angle EBC = 1/2 angle ABC.

Step 7: Let angle ABC = 2φ. Then angle ABE = angle EBC = φ.

Step 8: In triangle ABC, the sum of angles is 180°. So, angle BAC + angle ABC + angle ACB = 180°. Substituting angle BAC = 58°, and angle ABC = 2φ, angle ACB = 2θ. So, 58° + 2φ + 2θ = 180°, which simplifies to φ + θ = 61°.

Step 9: In triangle BFC, the sum of angles is 180°. So, angle BFC + angle FBC + angle FCB = 180°.

Step 10: Substitute angle FBC = φ and angle FCB = θ. So, angle BFC + φ + θ = 180°. From step 8, φ + θ = 61°, so angle BFC = 180° - 61° = 119°。

†Answer: 119 °

False-positive Rollout:

Step 1: Recognize that CD and BE are angle bisectors, so point F is the incenter of triangle ABC.

Step 2: Since CG is perpendicular to AB, triangle CGA is a right triangle.

Step 3: In triangle CGA, using the fact that the sum of angles is 180°, calculate angle CAG: angle CAG = 90° - angle ACG =90° - 32° = 58°.

Step 4: Angle BFC = 180° - angle CDE. CD is the angle bisector of angle ACB, angle ACD = angle BCD = 1/2 angle ACB.

Step 5: We have angle CDE = angle ACD = angle BCD = 1/2 angle ACB.

Step 6: Since angle ACB + angle ACG = 180°, angle ACB = 180° - 58° = 122°.

Step 7: Substitute angle ACB = 122°, angle ACD = 1/2 * 122° = 61°.

Step 8: Substitute angle ACD = 66°, angle BFC = 180° - 61° = 119°.

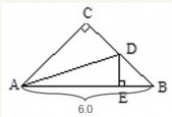
†Answer: 119 °

Analysis:

In this example, false-positive rollout leads to numerous instances of image-text perception inconsistency, and such behavior should not be encouraged in reasoning, even if it happens to yield the correct result.

Figure 20: False positive rollout analysis I.

Rollouts Analysis on Process (II)



Problem:
As shown in the figure, AC = BC, AD bisects angle CAB, then the perimeter of triangle DBE is () Choices: A:6cm B:7cm C:8cm D:9cm

Perfect Rollout:

Step 1: The angle bisector states that a point on an angle bisector is equidistant from the two sides of the angle. Since AD bisects angle CAB, DC ⊥ AC, DE ⊥ AB, we conclude that DC=DE.

Step 2: Since angle C = angle AED = 90°, angle CAD = angle EAD, we have triangle ACD ≅ triangle AED.

Step 3: We substitute AC with BC, so BC = AE.

Step 4: The perimeter of triangle DBE equals to DB + BE + DE, notice that AE + BE = AB

Step 5: From the diagram. AB = 6cm

Step 6: The answer is 6cm.

†Answer: A ✓

False-positive Rollout:

Step 1: Look at the figure and notice that AC = BC, so triangle ABC must be isosceles.

Step 2: AD is the angle bisector of angle CAB. Hmm, so it splits angle A into two equal parts.

Step 3: DE is perpendicular to AB.

Step 4: The question asks for the perimeter of triangle DBE. I need to find the lengths of DB, BE, and DE.

Step 5: Since ABC is isosceles with AC = BC, angles at A and B are equal.

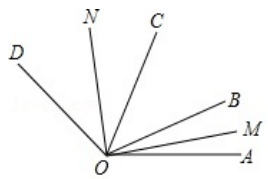
Step 6: Maybe triangle DBE is related to triangle ABC in some way.

Step 7: Let me guess that the perimeter of DBE is equal to AB. If AB is 6 cm, then the perimeter of DBE would be 6 cm. So the answer is A.

†Answer: A ✓

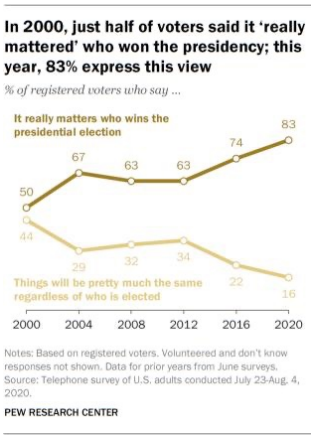
Analysis:
This type of learning pattern makes problems very difficult to solve when shortcuts in numerical and geometric relationships point in the wrong direction.

Figure 21: False positive rollout analysis II.



Question:
As shown in the figure: ∠AOB : ∠BOC : ∠COD = 2 : 3 : 4, ray OM and ON bisect ∠AOB and ∠COD respectively, and ∠MON = 84°, then ∠AOB is ( ).

Solution:
Step 1: Look at the diagram and note the angles ∠AOB, ∠BOC, and ∠COD are in the ratio 2:3:4.
Step 2: Recognize OM and ON bisect ∠AOB and ∠COD, creating smaller angles.
Step 3: Use the given ∠MON = 84° to set up an equation based on the ratios and angle bisectors.
Step 4: Solve the equation ∠AOB = 2 / (2 + 3 + 4) * 84° = 18.66°.
†Answer: ∠AOB is 18.66°.



Question:
As shown in the figure: ∠AOB : ∠BOC : ∠COD = 2 : 3 : 4, ray OM and ON bisect ∠AOB and ∠COD respectively, and ∠MON = 84°, then ∠AOB is ( ).

Solution:
Step 1: The dark brown line represents voters who think it really matters, showing an increasing trend from 50% in 2000 to 83% in 2020. The light brown line represents voters who think things will be much the same, showing a decreasing trend from 44% in 2000 to 16% in 2020.
Step 2: We need to know the least value of the light brown graph to the leftmost value of the dark brown graph.
Step 3: The ratio of the least value of the light brown graph and the leftmost value of the dark brown graph is 16:67.
†Answer: 16:67.

Figure 22: Bad case analysis on two process reward modeling variants.